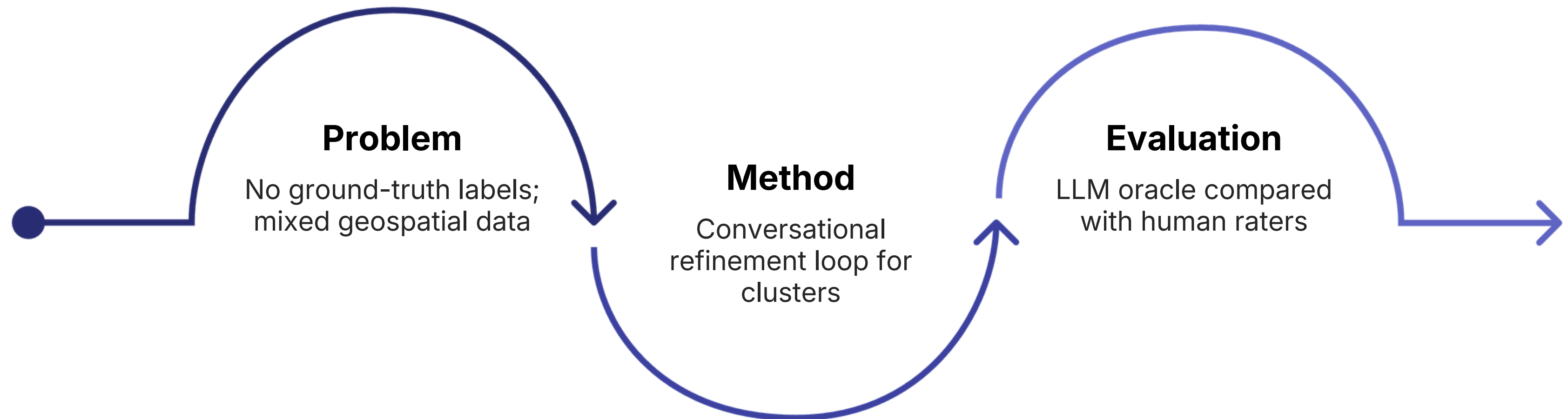


Conversational Clustering on UCDP GEDEvent

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Knowledge Discovery and Pattern Extraction — applying iterative, natural-language-driven cluster refinement to georeferenced conflict event data from the Russia-Ukraine theater.



Problem Statement

The Dataset

The **UCDP Georeferenced Event Dataset (GED)** contains structured, georeferenced records of organized violent events. Each record encodes:

- Event type
- Actor identities (side_a / side_b)
- Location: lat/lon + administrative divisions (adm1/adm2)
- Date and estimated casualties (best estimate of fatalities)

The Core Problem

The dataset has **no cluster labels**. Standard one-shot automated clustering (e.g. k-means) is highly sensitive to feature weighting and distance metric choices that users cannot easily specify upfront on a mixed structured dataset (numeric + categorical + geographic).

The Key Insight

A **conversational refinement loop** provides a channel to inject knowledge incrementally, turn by turn.

Research Questions & Hypotheses

RQ1 — Primary: Conversational Refinement vs. One-Shot

Does iterative conversational refinement produce measurably better clusters (by internal validity indices) than a single automated clustering run on GEDEvent data — and does the magnitude of improvement depend on the feature set used to represent events?

RQ2 — Secondary: LLM Oracle as Proxy for Human Judgment

Do cluster quality rankings produced by an LLM oracle correlate with human rater preferences, making oracle-driven evaluation a viable low-cost substitute for human rating?

Hypotheses

ID	Type	Statement
H1	Confirmatory	Conversational refinement improves Silhouette score vs. automated baseline (one-shot k-means), with 95% CI
H2	Exploratory	Improvement is larger with a richer feature set (event type + actor + location + casualties) than with location alone — refinement adds most value in high-dimensional, ambiguous feature spaces
H3	Confirmatory	Spearman's rank correlation between LLM oracle rankings and human rater rankings > 0.7 on a held-out sample of 20 cluster pairs

State of the Art — Six Research Threads

Thread 1 · Interactive & Conversational Clustering

- **Cohn et al. (EMNLP 2017)** — Active learning with must-link/cannot-link constraints; constraint vocabulary maps directly to anticipated NL instructions
- **Andrienko & Andrienko (2006)** — Interactive visual analytics on spatio-temporal data; motivates why users struggle to tune spatial vs. thematic weights by hand
- **Choo et al. — UTOPIAN (IEEE TVCG 2013)** — Iterative human feedback converges to more interpretable clusters than one-shot methods; empirical support for H1
- **Amershi et al. (AI Magazine 2014)** — Survey establishing vocabulary for interactive ML paradigms

Thread 2 · LLM-as-Evaluator

- **Zheng et al. — MT-Bench (arXiv 2023)** — Foundational paper on GPT-4 as proxy judge; sets our $\rho > 0.7$ bar. Key caveat: oracle biased toward its own output style
- **Guo et al. (arXiv 2023)** — Survey of LLM evaluation paradigms; anticipates failure modes (position bias, verbosity bias, cultural priors)
- **Chiang & Lee (ACL 2023)** — LLM ratings correlate moderately-to-highly in some settings, not others — motivates treating $\rho > 0.7$ as non-trivial

Thread 3 · Internal Cluster Validity Indices

- **Rousseeuw (1987)** — Defines Silhouette coefficient: reference-free, $[-1,1]$ interpretable, our primary metric
- **Davies & Bouldin (1979)** — DB index: penalizes compact-but-close clusters, a failure mode relevant to geographic event data
- **Arbelaitz et al. (Pattern Recognition 2013)** — Benchmarks 30 validity indices; Silhouette + DB most robust across mixed data types

Thread 4 · UCDP GED & Conflict Event Data

- **Sundberg & Melander (JPR 2013)** — Original GED methodological paper: coding rules, unit of analysis, known data quality issues (reporting bias, retroactive revision)
- **Croicu & Sundberg (2017)** — GED Codebook: authoritative source for all variable definitions; consulted for feature encodings

Thread 5 · Human-in-the-Loop ML

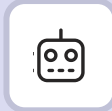
- **Settles (2012)** — Active learning theory; explains why conversational loop is epistemically different from passive automated tuning — each instruction is an information-efficient oracle query
- **Krippendorff (2004)** — Source for α inter-rater agreement statistic; handles ordinal ratings with more than two raters and missing data

Thread 6 · LLM-Enhanced Clustering ★ SOTA

- **Zhang et al. — ClusterLLM (EMNLP 2023)** — LLMs guide clustering via triplet evaluation; contrast: they use LLMs as a "guide" for math clustering; we test a fully conversational loop
- **Viswanathan et al. — Few-Shot Clustering (arXiv 2023)** — Single NL prompt generates thousands of pseudo-constraints, reducing human effort by 90%; directly validates our core assumption
- **Grootendorst — BERTopic (arXiv 2022)** — HDBSCAN + LLM post-hoc labeling; contrast: BERTopic does "LLM-labeling of geometric clusters"; we investigate "LLM-driven semantic clustering from the ground up"

Experimental Design

Three Conditions

Condition A — Baseline One-shot k-means, default features (location + event type), K by elbow method Role: Primary control	 Condition B — Conversational User refines clusters via natural language, up to 5 turns per session Role: Treatment	 Condition C — LLM Oracle LLM acts as the user in B; simulates many sessions cheaply Role: Proxy-validity arm
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Dataset Scope

- Russia-Ukraine conflict events, post-2022 (country codes UKR/RUS)
- Target sample: 1,000–3,000 events (downsampled with fixed seed if larger)
- Features: event_type, side_a/side_b actor names, adm1/adm2 region, lat/lon, fatalities estimate, date_start
- Encoding: one-hot for categoricals, min-max normalization for continuous, optional sentence embeddings for where_description

Outcome Measures

- **Primary:** Silhouette score (higher = better)
- **Secondary:** Davies-Bouldin index (lower = better); number of turns to stable assignment
- **For RQ2:** Spearman's ρ (oracle vs. human rankings); Krippendorff's α (inter-rater agreement)

Human Rating Protocol

- 5–8 raters, each rating 20 cluster pairs (forced-choice pairwise, within-subject)
- Shown: cluster label + 3 representative events per cluster
- Question: "Which grouping makes more sense as a meaningful category of conflict event?"

Statistical Plan

- 30 replications per condition, different random seeds
- Report Silhouette improvement distribution + 95% bootstrapped CI (H1)
- Power adequate to estimate Spearman's ρ with ± 0.15 precision
- Effect sizes and CIs reported honestly — no overclaiming

System Architecture

01

Data Layer

Input: UCDP GEDEvent 25.1 CSV → Filter UKR/RUS → one-hot + min-max normalization + optional embeddings → feature matrix + config

02

Clustering Engine

scikit-learn k-means with configurable K and feature set. Condition A: single automated run. Conditions B/C: initialized from same baseline, ready for refinement loop.

03

Conversational Loop

Up to 5 turns per session. Each turn: NL instruction → Claude API parses → updates clustering params → re-runs engine → logs instruction, interpretation, new params, new Silhouette score.

04

Evaluation & Logging

Writes cluster assignments + Silhouette + DB scores to JSONL after each turn.

Tech Stack

Python

Core language

scikit-learn

k-means + validity indices

sentence-transformers

where_description embeddings

Claude API

LLM interpreter + oracle

pandas + jsonlines

Data wrangling + logging